

Chapter 17: Complex Numbers, Powers & De Moivre's Formula

Concordia University · Department of Mechanical, Industrial & Aerospace Engineering

Course: Applied Ordinary Differential Equations (ENGR 213) | Textbook: Dennis G. Zill (7th Ed., Sections 17.1 & 17.2)

1. Executive Summary & First-Principles Philosophy

Complex numbers are the mathematical bridge that allows higher-order differential equations and vibrations to be solved elegantly. Without the imaginary unit $i = \sqrt{-1}$, oscillatory solutions like $\sin(\omega t)$ and $\cos(\omega t)$ would have to be handled through cumbersome trigonometric identities. Euler's formula $e^{i\theta} = \cos \theta + i \sin \theta$ unifies algebraic exponentials with oscillatory trigonometry.

2. Core Mechanics & Mathematical Engine

A. Rectangular and Polar Representations

- Rectangular Form:** $z = x + iy$
- Modulus:** $|z| = r = \sqrt{x^2 + y^2}$
- Argument:** $\theta = \arg(z) = \text{atan2}(y, x)$ (Principal argument: $-\pi < \text{Arg}(z) \leq \pi$)
- Polar / Exponential Form:**

$$z = r(\cos \theta + i \sin \theta) = re^{i\theta}$$

B. Powers & De Moivre's Formula

For any integer n :

$$z^n = (re^{i\theta})^n = r^n e^{in\theta} = r^n (\cos(n\theta) + i \sin(n\theta))$$

C. Roots of Complex Numbers (n -th Roots)

To solve $w^n = z_0 = r_0 e^{i\theta_0}$, there exist exactly n distinct roots distributed symmetrically on a circle of radius $r_0^{1/n}$ with angular spacing $\frac{2\pi}{n}$:

$$w_k = r_0^{1/n} \exp\left(i \frac{\theta_0 + 2k\pi}{n}\right), \quad k = 0, 1, 2, \dots, n-1$$

3. Curriculum-Grounded Visual Reference

Addition: $z_1 + z_2 = (x_1 + iy_1) + (x_2 + iy_2) = (x_1 + x_2) + i(y_1 + y_2)$

Subtraction: $z_1 - z_2 = (x_1 + iy_1) - (x_2 + iy_2) = (x_1 - x_2) + i(y_1 - y_2)$

Multiplication: $z_1 \cdot z_2 = (x_1 + iy_1)(x_2 + iy_2)$
 $= x_1x_2 - y_1y_2 + i(y_1x_2 + x_1y_2)$

Division: $\frac{z_1}{z_2} = \frac{x_1 + iy_1}{x_2 + iy_2}$
 $= \frac{x_1x_2 + y_1y_2}{x_2^2 + y_2^2} + i \frac{y_1x_2 - x_1y_2}{x_2^2 + y_2^2}$

Figure 17.1: The Argand complex plane: Cartesian coordinates (x, y) vs Polar (r, θ) — from Zill 7th Ed. Chapter 17 (Fig. 17.1.1).

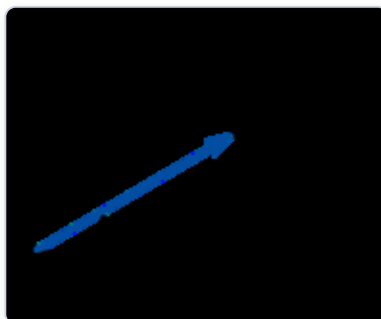


Figure 17.2: Roots of a complex number evenly spaced on a circle in the complex plane — from Zill 7th Ed. Chapter 17 (Fig. 17.2.1).

4. Fully Worked Exam Archetype: Finding All Cube Roots

Problem: Find all cube roots of $z = -8i$ in exact rectangular form.

Step-by-Step Solution:

- **Step 1: Convert z to polar form:** Modulus $r = |-8i| = 8$. Argument: $-8i$ lies on the negative imaginary axis $\implies \theta = -\frac{\pi}{2}$ (or $\frac{3\pi}{2}$).

$$z = 8e^{-i\pi/2}$$

- **Step 2: Apply the n -th Root Formula for $n = 3$:**

$$w_k = 8^{1/3} \exp\left(i \frac{-\pi/2 + 2k\pi}{3}\right) = 2 \exp\left(i \left[-\frac{\pi}{6} + \frac{2k\pi}{3}\right]\right), \quad k = 0, 1, 2$$

- **Step 3: Evaluate each root:**

- **For** $k = 0$:

$$\theta_0 = -\frac{\pi}{6} \implies w_0 = 2 \left(\cos \left(-\frac{\pi}{6} \right) + i \sin \left(-\frac{\pi}{6} \right) \right) = 2 \left(\frac{\sqrt{3}}{2} - \frac{1}{2}i \right) = \sqrt{3} - i$$

- **For** $k = 1$:

$$\theta_1 = -\frac{\pi}{6} + \frac{2\pi}{3} = \frac{\pi}{2} \implies w_1 = 2 \left(\cos \left(\frac{\pi}{2} \right) + i \sin \left(\frac{\pi}{2} \right) \right) = 2(0 + i) = 2i$$

- **For** $k = 2$:

$$\theta_2 = -\frac{\pi}{6} + \frac{4\pi}{3} = \frac{7\pi}{6} \implies w_2 = 2 \left(\cos \left(\frac{7\pi}{6} \right) + i \sin \left(\frac{7\pi}{6} \right) \right) = 2 \left(-\frac{\sqrt{3}}{2} - \frac{1}{2}i \right) = -\sqrt{3} - i$$

5. Rapid Exam Traps & Red Flags

-  **Trap 1: Quadrant Checks in** $\arctan(y/x)$: Never blindly calculate $\arctan(y/x)$! If $z = -1 - i$, $y/x = 1$, but $\arctan(1) = \pi/4$ (Quadrant I), whereas z is in Quadrant III. You must add π (or subtract π) to get $-\frac{3\pi}{4}$.
-  **Trap 2: Forgetting Root Spacing**: The roots are separated by exactly $\frac{2\pi}{n}$ radians. If your roots are not symmetrically distributed on a circle, your angular arithmetic has an error.